EVIDENCIAS EJECUCIÓN LABORATORIO 2

En la siguiente imagen se puede observar la salida del paso 2:

- LeaderId y LeaderEpoch.

```
sorjuelav@DESKTOP-7O817GL:/home/equipo6/kafka_env$ sudo script laboratorio_2.txt
Script started, output log file is 'laboratorio_2.txt'.
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker compose -f docker-compose-lab-kafka.yml ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker compose -f docker-compose-lab-kafka.yml up -d
[+] up 5/5
 ✔Container broker3          Healthy                                                                                                  19.1s
 ✔Container broker2          Healthy                                                                                                  19.1s
 ✔Container broker1          Healthy                                                                                                  19.1s
 ✔Container kafka-ui         Started                                                                                                   0.4s
 ✔Container bootstrap-topics Started                                                                                                   0.3s
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker compose -f docker-compose-lab-kafka.yml ps
NAME       IMAGE                             COMMAND                  SERVICE    CREATED          STATUS                    PORTS
broker1    apache/kafka:3.9.0                "/__cacert_entrypoin…"   broker1    58 minutes ago   Up 26 seconds (healthy)   0.0.0.0:9092->9092/tcp, [::]
:9092->9092/tcp, 0.0.0.0:9101->9101/tcp, [::]:9101->9101/tcp
broker2    apache/kafka:3.9.0                "/__cacert_entrypoin…"   broker2    58 minutes ago   Up 26 seconds (healthy)   0.0.0.0:9094->9094/tcp, [::]
:9094->9094/tcp, 0.0.0.0:9102->9102/tcp, [::]:9102->9102/tcp
broker3    apache/kafka:3.9.0                "/__cacert_entrypoin…"   broker3    58 minutes ago   Up 27 seconds (healthy)   0.0.0.0:9096->9096/tcp, [::]
:9096->9096/tcp, 0.0.0.0:9103->9103/tcp, [::]:9103->9103/tcp
kafka-ui   ghcr.io/kafbat/kafka-ui:v1.1.0    "/bin/sh -c 'java --…"   kafka-ui   58 minutes ago   Up 26 seconds             0.0.0.0:8081->8080/tcp, [::]
:8081->8080/tcp
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kq describe --status
kq: command not found
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kt='docker exec broker1 /opt/kafka/bin/kafka-topics.sh --bootstrap-server broker1:19092'
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kq='docker exec broker1 /opt/kafka/bin/kafka-metadata-quorum.sh --bootstrap-server broker1:19092'
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kq describe --status
kq: command not found
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kq describe --status
kq: command not found
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# alias kq='docker exec broker1 /opt/kafka/bin/kafka-metadata-quorum.sh --bootstrap-server broker1:19092'
alias kt='docker exec broker1 /opt/kafka/bin/kafka-topics.sh --bootstrap-server broker1:19092'
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kq describe --status
ClusterId:              dian-kafka-lab-2026
LeaderId:               2
LeaderEpoch:            162
HighWatermark:          7655
MaxFollowerLag:         0
MaxFollowerLagTimeMs:   -2458
CurrentVoters:          [{"id": 1, "directoryId": null, "endpoints": ["CONTROLLER://broker1:9093"]}, {"id": 2, "directoryId": null, "endpoints": ["CONT
ROLLER://broker2:9093"]}, {"id": 3, "directoryId": null, "endpoints": ["CONTROLLER://broker3:9093"]}]
CurrentObservers:       []
root@DESKTOP-7O817GL:/home/equipo6/kafka_env#
```

```
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kq describe --replication
NodeId  DirectoryId            LogEndOffset    Lag     LastFetchTimestamp      LastCaughtUpTimestamp   Status
1       AAAAAAAAAAAAAAAAAAAAAA  7894            0       1792468489195           1792468489195           Leader
2       AAAAAAAAAAAAAAAAAAAAAA  7885            9       1792468487114           1792468487114           Follower
3       AAAAAAAAAAAAAAAAAAAAAA  7885            9       1792468487114           1792468487114           Follower
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker exec broker1 ls -lh /var/lib/kafka/data/__cluster_metadata-0/
total 592K
-rw-r--r-- 1 appuser appuser  10M Oct 20 03:55 00000000000000000000.index
-rw-r--r-- 1 appuser appuser 552K Oct 20 03:55 00000000000000000000.log
-rw-r--r-- 1 appuser appuser  10M Oct 20 03:55 00000000000000000000.timeindex
-rw-r--r-- 1 appuser appuser   10 Oct 20 03:51 00000000000000007183.snapshot
-rw-r--r-- 1 appuser appuser   10 Oct 20 03:52 00000000000000007535.snapshot
-rw-r--r-- 1 appuser appuser   10 Oct 20 03:52 00000000000000007595.snapshot
-rw-r--r-- 1 appuser appuser   10 Oct 20 03:54 00000000000000007885.snapshot
-rw-r--r-- 1 appuser appuser  634 Oct 20 03:55 leader-epoch-checkpoint
-rw-r--r-- 1 appuser appuser   43 Oct 20 02:53 partition.metadata
-rw-r--r-- 1 appuser appuser  155 Oct 20 03:55 quorum-state
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker exec broker1 /opt/kafka/bin/kafka-dump-log.sh --cluster-metadata-decoder \
  --files /var/lib/kafka/data/__cluster_metadata-0/00000000000000000000.log | grep -E "TOPIC_RECORD|PARTITION_RECORD|REGISTER_BROKER" | head -20
| offset: 5 CreateTime: 1792464795594 keySize: -1 valueSize: 144 sequence: -1 headerKeys: [] payload: {"type":"REGISTER_BROKER_RECORD","version":3,"dat
a":{"brokerId":3,"isMigratingZkBroker":false,"incarnationId":"gKjivckgQQ-WLOM9Vrb5Cw","brokerEpoch":5,"endPoints":[{"name":"INTERNAL","host":"broker3",
"port":19092,"securityProtocol":0},{"name":"EXTERNAL","host":"localhost","port":9096,"securityProtocol":0}],"features":[{"name":"kraft.version","minSup
portedVersion":0,"maxSupportedVersion":1},{"name":"metadata.version","minSupportedVersion":1,"maxSupportedVersion":21}],"rack":null,"fenced":true,"inCo
ntrolledShutdown":false,"logDirs":["qKp3wcYKNg7Ujx49xm6pEA"]}}
| offset: 6 CreateTime: 1792464795594 keySize: -1 valueSize: 144 sequence: -1 headerKeys: [] payload: {"type":"REGISTER_BROKER_RECORD","version":3,"dat
a":{"brokerId":1,"isMigratingZkBroker":false,"incarnationId":"dmUIVn80T3mbatBUCsLb7Q","brokerEpoch":6,"endPoints":[{"name":"INTERNAL","host":"broker1",
"port":19092,"securityProtocol":0},{"name":"EXTERNAL","host":"localhost","port":9092,"securityProtocol":0}],"features":[{"name":"kraft.version","minSup
portedVersion":0,"maxSupportedVersion":1},{"name":"metadata.version","minSupportedVersion":1,"maxSupportedVersion":21}],"rack":null,"fenced":true,"inCo
ntrolledShutdown":false,"logDirs":["IiXMEC4kdZ6PKukgvq58fw"]}}
| offset: 7 CreateTime: 1792464795594 keySize: -1 valueSize: 144 sequence: -1 headerKeys: [] payload: {"type":"REGISTER_BROKER_RECORD","version":3,"dat
a":{"brokerId":2,"isMigratingZkBroker":false,"incarnationId":"N3uDxS3sTayiit3BaDrRlg","brokerEpoch":7,"endPoints":[{"name":"INTERNAL","host":"broker2",
"port":19092,"securityProtocol":0},{"name":"EXTERNAL","host":"localhost","port":9094,"securityProtocol":0}],"features":[{"name":"kraft.version","minSup
portedVersion":0,"maxSupportedVersion":1},{"name":"metadata.version","minSupportedVersion":1,"maxSupportedVersion":21}],"rack":null,"fenced":true,"inCo
ntrolledShutdown":false,"logDirs":["RRRE47Pear5ZXIpFrB48WA"]}}
| offset: 41 CreateTime: 1792464806704 keySize: -1 valueSize: 45 sequence: -1 headerKeys: [] payload: {"type":"TOPIC_RECORD","version":0,"data":{"name"
:"dian.facturacion.emitida","topicId":"GKi2jQA9Ruy2GTEhZHzkTw"}}
| offset: 44 CreateTime: 1792464806704 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":{"
partitionId":0,"topicId":"GKi2jQA9Ruy2GTEhZHzkTw","replicas":[2,3,1],"isr":[2,3,1],"removingReplicas":[],"addingReplicas":[],"leader":2,"leaderEpoch":0
,"partitionEpoch":0,"directories":["RRRE47Pear5ZXIpFrB48WA","qKp3wcYKNg7Ujx49xm6pEA","IiXMEC4kdZ6PKukgvq58fw"]}}
| offset: 45 CreateTime: 1792464806704 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":{"
partitionId":1,"topicId":"GKi2jQA9Ruy2GTEhZHzkTw","replicas":[3,1,2],"isr":[3,1,2],"removingReplicas":[],"addingReplicas":[],"leader":3,"leaderEpoch":0
,"partitionEpoch":0,"directories":["qKp3wcYKNg7Ujx49xm6pEA","IiXMEC4kdZ6PKukgvq58fw","RRRE47Pear5ZXIpFrB48WA"]}}
| offset: 46 CreateTime: 1792464806704 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":{"
partitionId":2,"topicId":"GKi2jQA9Ruy2GTEhZHzkTw","replicas":[1,2,3],"isr":[1,2,3],"removingReplicas":[],"addingReplicas":[],"leader":1,"leaderEpoch":0
,"partitionEpoch":0,"directories":["IiXMEC4kdZ6PKukgvq58fw","RRRE47Pear5ZXIpFrB48WA","qKp3wcYKNg7Ujx49xm6pEA"]}}
| offset: 47 CreateTime: 1792464806704 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":{"
partitionId":3,"topicId":"GKi2jQA9Ruy2GTEhZHzkTw","replicas":[1,2,3],"isr":[1,2,3],"removingReplicas":[],"addingReplicas":[],"leader":1,"leaderEpoch":0
,"partitionEpoch":0,"directories":["IiXMEC4kdZ6PKukgvq58fw","RRRE47Pear5ZXIpFrB48WA","qKp3wcYKNg7Ujx49xm6pEA"]}}
| offset: 48 CreateTime: 1792464806704 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":{"
partitionId":4,"topicId":"GKi2jQA9Ruy2GTEhZHzkTw","replicas":[2,3,1],"isr":[2,3,1],"removingReplicas":[],"addingReplicas":[],"leader":2,"leaderEpoch":0
```

En la siguiente imagen se puede observar la salida del paso 4:

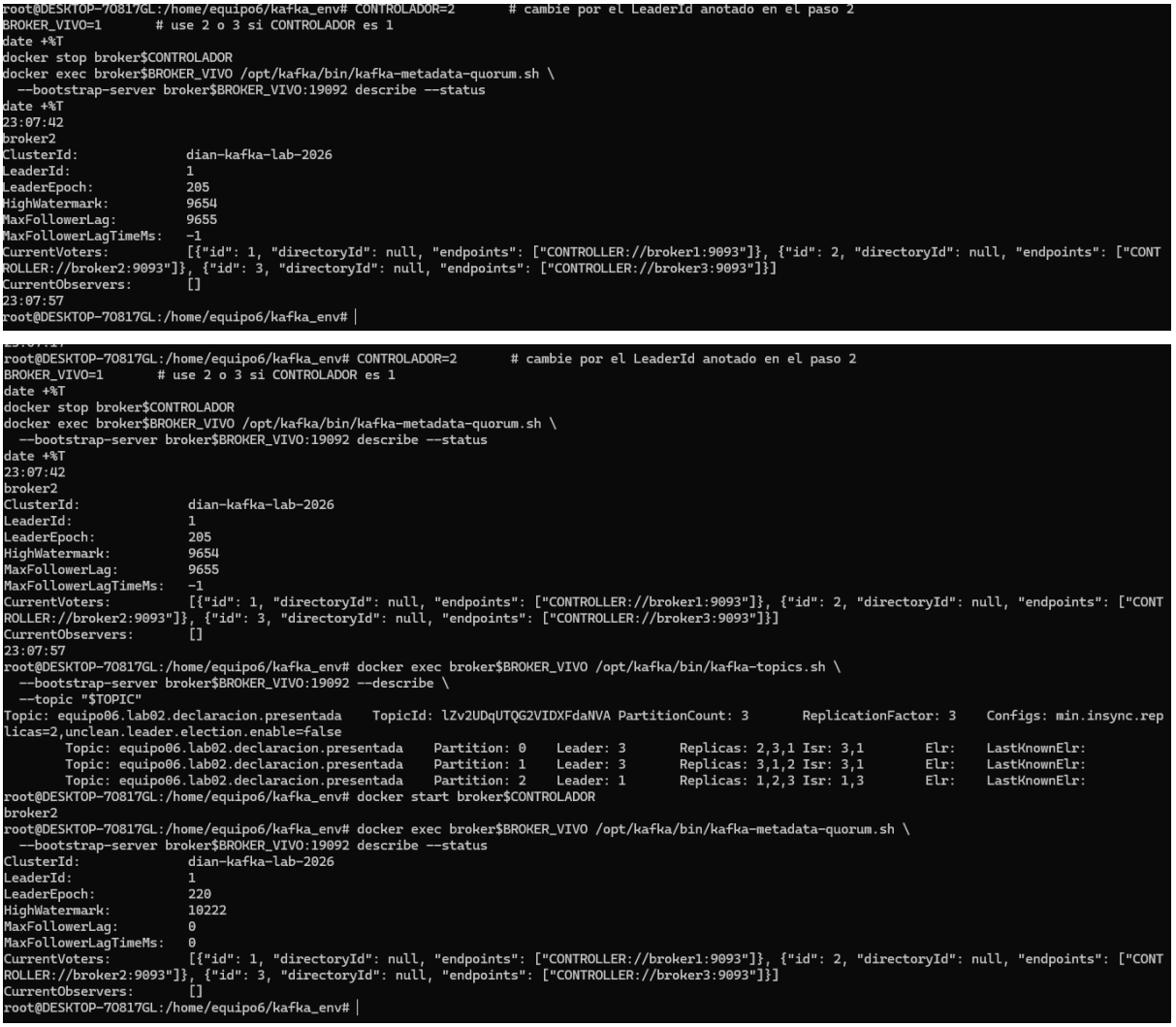
- Registros del paso 4 (TOPIC_RECORD y un PARTITION_RECORD).



```
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# EQUIPO=equipo06
TOPIC=$EQUIPO.lab02.declaracion.presentada
kt --create --topic "$TOPIC" --partitions 3 --replication-factor 3
docker exec broker1 /opt/kafka/bin/kafka-dump-log.sh --cluster-metadata-decoder \
  --files /var/lib/kafka/data/__cluster_metadata-0/00000000000000000000.log | grep -A3 "$TOPIC" | head -20
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, bu
t not both.
Created topic equipo06.lab02.declaracion.presentada.
| offset: 8654 CreateTime: 1792468851735 keySize: -1 valueSize: 58 sequence: -1 headerKeys: [] payload: {"type":"TOPIC_RECORD","version":0,"data":{"nam
e":"equipo06.lab02.declaracion.presentada","topicId":"lZv2UDqUTQG2VIDXFdaNVA"}}
| offset: 8655 CreateTime: 1792468851735 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":
{"partitionId":0,"topicId":"lZv2UDqUTQG2VIDXFdaNVA","replicas":[2,3,1],"isr":[2,3,1],"removingReplicas":[],"addingReplicas":[],"leader":2,"leaderEpoch"
:0,"partitionEpoch":0,"directories":["RRRE47Pear5ZXIpFrB48WA","qKp3wcYKNg7Ujx49xm6pEA","IiXMEC4kdZ6PKukgvq58fw"]}}
| offset: 8656 CreateTime: 1792468851735 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":
{"partitionId":1,"topicId":"lZv2UDqUTQG2VIDXFdaNVA","replicas":[3,1,2],"isr":[3,1,2],"removingReplicas":[],"addingReplicas":[],"leader":3,"leaderEpoch"
:0,"partitionEpoch":0,"directories":["qKp3wcYKNg7Ujx49xm6pEA","IiXMEC4kdZ6PKukgvq58fw","RRRE47Pear5ZXIpFrB48WA"]}}
| offset: 8657 CreateTime: 1792468851735 keySize: -1 valueSize: 113 sequence: -1 headerKeys: [] payload: {"type":"PARTITION_RECORD","version":1,"data":
{"partitionId":2,"topicId":"lZv2UDqUTQG2VIDXFdaNVA","replicas":[1,2,3],"isr":[1,2,3],"removingReplicas":[],"addingReplicas":[],"leader":1,"leaderEpoch"
:0,"partitionEpoch":0,"directories":["IiXMEC4kdZ6PKukgvq58fw","RRRE47Pear5ZXIpFrB48WA","qKp3wcYKNg7Ujx49xm6pEA"]}}
root@DESKTOP-7O817GL:/home/equipo6/kafka_env#
```

En la siguiente imagen se puede observar la salida del paso 5:

- Caída y elección del controlador



```
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# CONTROLADOR=2       # cambie por el LeaderId anotado en el paso 2
BROKER_VIVO=1       # use 2 o 3 si CONTROLADOR es 1
date +%T
docker stop broker$CONTROLADOR
docker exec broker$BROKER_VIVO /opt/kafka/bin/kafka-metadata-quorum.sh \
  --bootstrap-server broker$BROKER_VIVO:19092 describe --status
date +%T
23:07:42
broker2
ClusterId:              dian-kafka-lab-2026
LeaderId:               1
LeaderEpoch:            205
HighWatermark:          9654
MaxFollowerLag:         9655
MaxFollowerLagTimeMs:   -1
CurrentVoters:          [{"id": 1, "directoryId": null, "endpoints": ["CONTROLLER://broker1:9093"]}, {"id": 2, "directoryId": null, "endpoints": ["CONT
ROLLER://broker2:9093"]}, {"id": 3, "directoryId": null, "endpoints": ["CONTROLLER://broker3:9093"]}]
CurrentObservers:       []
23:07:57
root@DESKTOP-7O817GL:/home/equipo6/kafka_env#
```



```
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# CONTROLADOR=2       # cambie por el LeaderId anotado en el paso 2
BROKER_VIVO=1       # use 2 o 3 si CONTROLADOR es 1
date +%T
docker stop broker$CONTROLADOR
docker exec broker$BROKER_VIVO /opt/kafka/bin/kafka-metadata-quorum.sh \
  --bootstrap-server broker$BROKER_VIVO:19092 describe --status
date +%T
23:07:42
broker2
ClusterId:              dian-kafka-lab-2026
LeaderId:               1
LeaderEpoch:            205
HighWatermark:          9654
MaxFollowerLag:         9655
MaxFollowerLagTimeMs:   -1
CurrentVoters:          [{"id": 1, "directoryId": null, "endpoints": ["CONTROLLER://broker1:9093"]}, {"id": 2, "directoryId": null, "endpoints": ["CONT
ROLLER://broker2:9093"]}, {"id": 3, "directoryId": null, "endpoints": ["CONTROLLER://broker3:9093"]}]
CurrentObservers:       []
23:07:57
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker exec broker$BROKER_VIVO /opt/kafka/bin/kafka-topics.sh \
  --bootstrap-server broker$BROKER_VIVO:19092 --describe \
  --topic "$TOPIC"
Topic: equipo06.lab02.declaracion.presentada    TopicId: lZv2UDqUTQG2VIDXFdaNVA PartitionCount: 3       ReplicationFactor: 3    Configs: min.insync.rep
licas=2,unclean.leader.election.enable=false
        Topic: equipo06.lab02.declaracion.presentada    Partition: 0    Leader: 3       Replicas: 2,3,1 Isr: 3,1        Elr:    LastKnownElr:
        Topic: equipo06.lab02.declaracion.presentada    Partition: 1    Leader: 3       Replicas: 3,1,2 Isr: 3,1        Elr:    LastKnownElr:
        Topic: equipo06.lab02.declaracion.presentada    Partition: 2    Leader: 1       Replicas: 1,2,3 Isr: 1,3        Elr:    LastKnownElr:
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker start broker$CONTROLADOR
broker2
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker exec broker$BROKER_VIVO /opt/kafka/bin/kafka-metadata-quorum.sh \
  --bootstrap-server broker$BROKER_VIVO:19092 describe --status
ClusterId:              dian-kafka-lab-2026
LeaderId:               1
LeaderEpoch:            220
HighWatermark:          10222
MaxFollowerLag:         0
MaxFollowerLagTimeMs:   0
CurrentVoters:          [{"id": 1, "directoryId": null, "endpoints": ["CONTROLLER://broker1:9093"]}, {"id": 2, "directoryId": null, "endpoints": ["CONT
ROLLER://broker2:9093"]}, {"id": 3, "directoryId": null, "endpoints": ["CONTROLLER://broker3:9093"]}]
CurrentObservers:       []
root@DESKTOP-7O817GL:/home/equipo6/kafka_env#
```

En la siguiente imagen se puede observar la salida del paso 6:

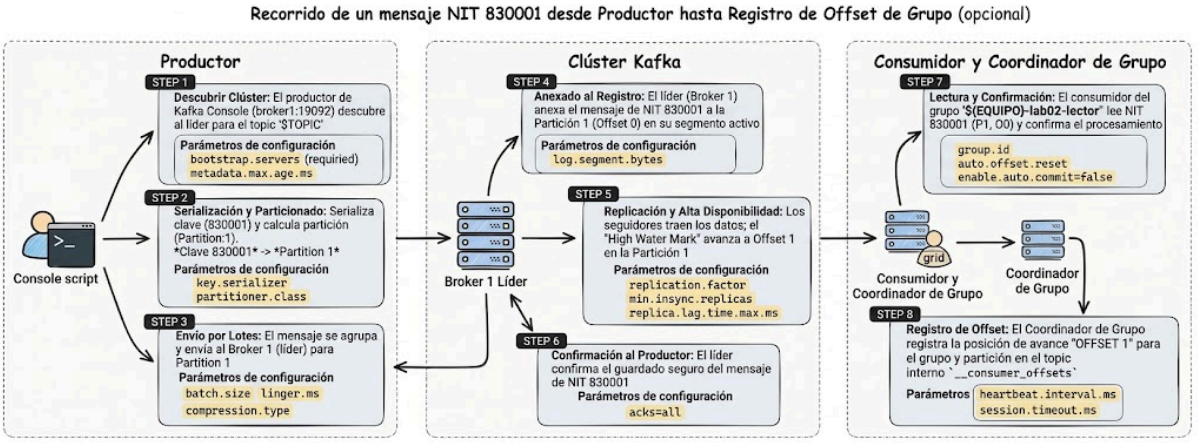
- Producir y consumir registrando offsets

```
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# printf "830001|{\"nit\":\"830001\",\"formulario\":\"110\",\"periodo\":\"2026-08\"}\n830001|{\"nit\":\"830
001\",\"formulario\":\"300\",\"periodo\":\"2026-08\"}\n900123|{\"nit\":\"900123\",\"formulario\":\"110\",\"periodo\":\"2026-08\"}\n" | \
docker exec -i broker1 /opt/kafka/bin/kafka-console-producer.sh --bootstrap-server broker1:19092 \
  --topic "$TOPIC" --property parse.key=true --property key.separator="|"
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker exec broker1 /opt/kafka/bin/kafka-console-consumer.sh --bootstrap-server broker1:19092 \
  --topic "$TOPIC" --from-beginning --timeout-ms 8000 \
  --property print.key=true --property print.partition=true --property print.offset=true 2>/dev/null
Partition:1     Offset:0        900123  {"nit":"900123","formulario":"110","periodo":"2026-08"}
Partition:2     Offset:0        830001  {"nit":"830001","formulario":"110","periodo":"2026-08"}
Partition:2     Offset:1        830001  {"nit":"830001","formulario":"300","periodo":"2026-08"}
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# docker exec broker1 /opt/kafka/bin/kafka-console-consumer.sh --bootstrap-server broker1:19092 \
  --topic "$TOPIC" --from-beginning --group "${EQUIPO}-lab02-lector" --timeout-ms 8000 2>/dev/null >/dev/null
docker exec broker1 /opt/kafka/bin/kafka-consumer-groups.sh --bootstrap-server broker1:19092 --describe --group "${EQUIPO}-lab02-lector"

Consumer group 'equipo06-lab02-lector' has no active members.

GROUP                TOPIC                                PARTITION  CURRENT-OFFSET  LOG-END-OFFSET  LAG             CONSUMER-ID     HOST
CLIENT-ID
equipo06-lab02-lector equipo06.lab02.declaracion.presentada 1          1               1               0               -               -
-
equipo06-lab02-lector equipo06.lab02.declaracion.presentada 0          0               0               0               -               -
-
equipo06-lab02-lector equipo06.lab02.declaracion.presentada 2          2               2               0               -               -
-root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kt --list | grep consumer_offsets
__consumer_offsets
root@DESKTOP-7O817GL:/home/equipo6/kafka_env# kt --list | grep consumer_offsets
__consumer_offsets
root@DESKTOP-7O817GL:/home/equipo6/kafka_env#
```

1. El diagrama del paso 7.



2. Un párrafo: ¿por qué KRaft simplifica la operación frente a ZooKeeper en una entidad como la DIAN?
    a. Se elimina la necesidad de instalar, configurar y mantener un sistema externo. Al unificar todo bajo la misma arquitectura de Kafka, se reducen los costos de infraestructura, la administración, la operación, etc. El sistema responde mucho más rápido ante fallas y escala de manera eficiente. En el ámbito corporativo de una entidad pública operar a menor costo posible sistemas robustos es elemental para el país.